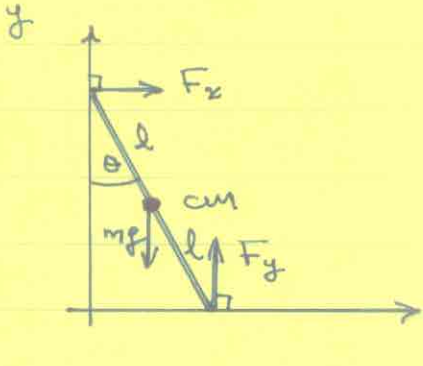


Sliding Ladder



- Contacts are frictionless
- θ completely describes the configuration
- Mass m
- Length $= 2l$

The contact forces F_x and F_y are normal to the walls because of the absence of friction.

$$I(\text{about cm}) = \int_{-l}^l dx \cdot \frac{m}{2l} \cdot x^2 = \frac{m}{2l} \frac{1}{3} x^3 \Big|_{-l}^l = \frac{m}{3l} l^3 = \underline{\underline{\frac{1}{3} m l^2}}$$

$$E = KE(\text{cm}) + KE(\text{rotation}) + \text{Pot}(\text{gravity})$$

$$x_{\text{cm}} = l \sin \theta; \quad y_{\text{cm}} = l \cos \theta$$

$$\dot{x}_{\text{cm}} = l \cos \theta \dot{\theta}; \quad \dot{y}_{\text{cm}} = -l \sin \theta \dot{\theta}$$

$$KE(\text{cm}) = \frac{1}{2} m [c^2 \theta \dot{\theta}^2 + s^2 \theta \dot{\theta}^2] l^2 = \frac{1}{2} m l^2 \dot{\theta}^2$$

$$KE(\text{rot}) = \frac{1}{2} I \dot{\theta}^2 = \frac{1}{2} \cdot \frac{1}{3} m l^2 \dot{\theta}^2 = \frac{1}{6} m l^2 \dot{\theta}^2$$

$$P(\text{grav}) = m g y = m g l \cos \theta$$

$$\Rightarrow E = \frac{2}{3} m l^2 \dot{\theta}^2 + mgl \cos \theta$$

The acceleration of the CM is:

$$F_{x/m} = \ddot{x}_{cm} = -l \sin \theta \dot{\theta}^2 + l \cos \theta \ddot{\theta}$$

$$F_{y/m} = \ddot{y}_{cm} = -l \cos \theta \dot{\theta}^2 - l \sin \theta \ddot{\theta}$$

only in equilibrium (?)

(= -g)

(no, always true)

Let the ladder start from rest at $\theta = 0$ ($\dot{\theta} = 0$)

$$\Rightarrow E = mgl = \frac{2}{3} m l^2 \dot{\theta}^2 + mgl \cos \theta$$

$$\Rightarrow \dot{\theta}^2 = \frac{3}{2} \frac{g}{l} (1 - \cos \theta)$$

$$\Rightarrow \ddot{\theta} = \frac{3}{4} \frac{g}{l} \sin \theta$$

The ladder loses contact with the vertical wall when $F_x = 0$:

$$0 = -l \left[\frac{3}{2} \frac{g}{l} (1 - \cos \theta) \right] \sin \theta + l \left[\frac{3}{4} \frac{g}{l} \sin \theta \right] \cos \theta$$

$$\Rightarrow 0 = -(1 - \cos \theta) + \frac{1}{2} \cos \theta = -1 + \cos \theta + \frac{1}{2} \cos \theta$$

$$\Rightarrow \underline{\underline{\cos \theta = \frac{2}{3}}} \Rightarrow \underline{\underline{\theta = 48.2^\circ}}$$